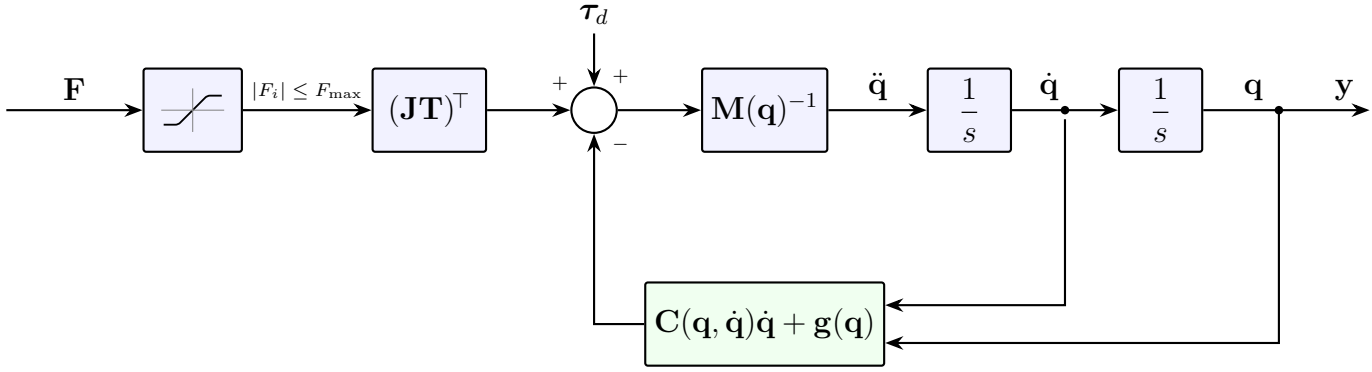

$\boldsymbol{\tau}_d$
$\mathbf{F}$
$|F_i| \leq F_{\max}$
$(\mathbf{JT})^\top$
+
+
−
$\mathbf{M}(\mathbf{q})^{-1}$
$\ddot{\mathbf{q}}$
$\frac{1}{s}$
$\dot{\mathbf{q}}$
$\frac{1}{s}$
$\mathbf{q}$
$\mathbf{y}$
$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q})$